Adaptive Trial Design for the RAS-Inhibitor Era: Combinations, Resistance, and Biomarker-Guided Therapy

Naim Rashid, PhD

Department of Biostatistics · Lineberger Comprehensive Cancer Center · UNC

Today’s argument

> **New RAS therapies in PDAC need new kinds of trials.**

Methods to optimize combination doses are established. Designs that adapt to emerging resistance are enrolling.

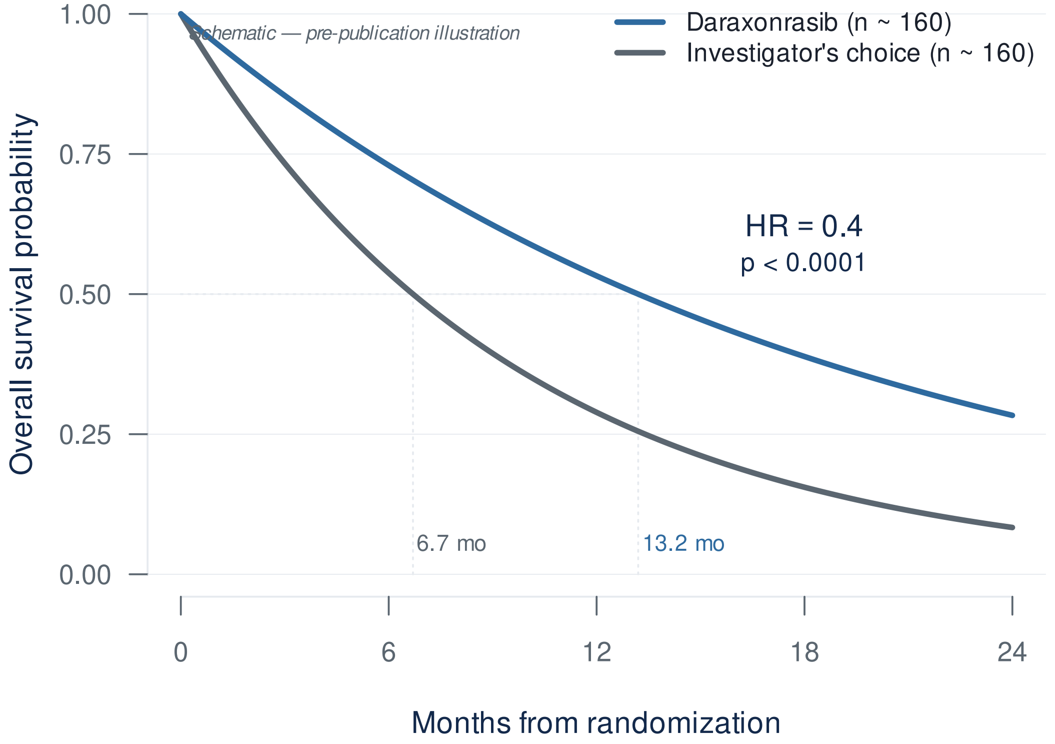
> **The difficulty: getting them calibrated and launched.**

The clinical case

Why combinations, and why now?

The RAS revolution has arrived for PDAC

- **RASolute 302** (2L mPDAC, *NEJM* 2026)[1]:
  - **Median OS: 13.2 vs 6.7 months**
  - **HR 0.40 — 60% reduction in risk of death** (p < 0.0001)
- Daraxonrasib (RMC-6236) — oral, multi-selective RAS(ON) inhibitor
- ASCO 2026 Plenary, May 31 (Wolpin)



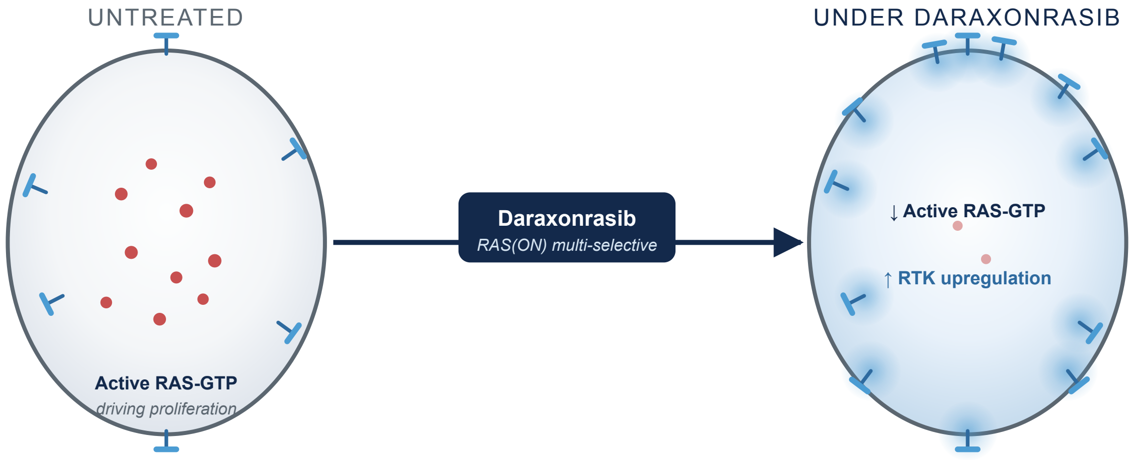
…but monotherapy has visible ceilings

RMC-6236-001 (PDAC cohort, NEJM 2026)[2]

> **Half of patients on monotherapy at 300 mg already need dose adjustments.** Combinations layer on top — with less room to spare.

Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface — the mechanistic case for biomarker-driven, resistance-preemptive combinations[3]
- **1L combination cohort** (daraxonrasib **200 mg** + GnP, n=40)[4]:
  - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**



RTK upregulation is simultaneously a resistance mechanism and a rational combination target

The dose-finding problem

Why MTD only may not be best for RAS combinations, and what replaces it.

Why combinations break the rules you grew up on

- **Rivière's review of 162 combination Phase I trials: 88% used 3+3** even when both drugs were escalated[5,6].
- **Why 3+3 doesn't fit combinations.** Built for single-agent cytotoxics[7], it assumes monotonic dose-toxicity, Cycle-1-observable DLTs, and a single dose dimension — combinations strain all three.
- **The two-drug dose map is never built.** In practice: pick A's dose, fix it, add B. Two monotherapy escalations run in series; the joint efficacy/toxicity behavior across (A, B) pairs (the "joint surface") is never characterized.

Two-drug dose map: how efficacy and toxicity change across every (A, B) dose pair — not just along one drug's axis.

The Amgen case: sotorasib + pembrolizumab

Same sponsor, two different designs:

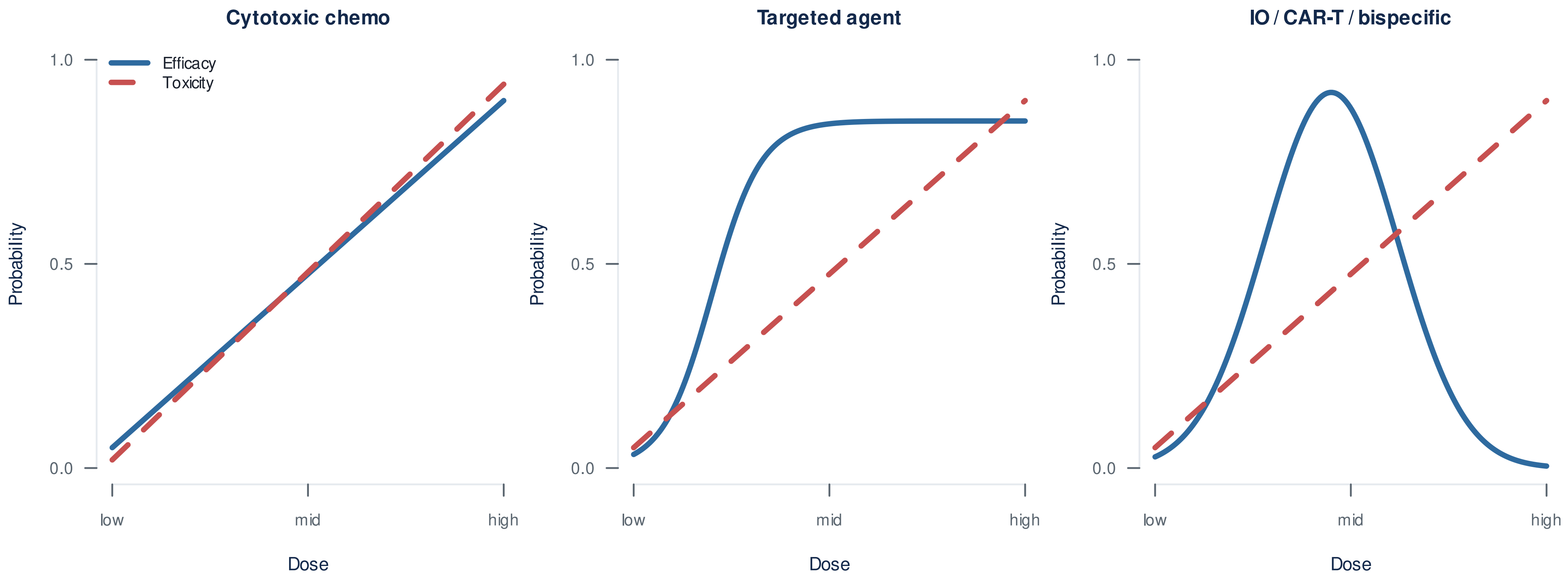
- **Monotherapy (CodeBreaK 100):** Bayesian model-based, single-agent surface
- **Combination:** IO dose fixed, only sotorasib varied — joint surface never modeled

The defining toxicity:

- Severe **immune-mediated** hepatotoxicity, not predictable from the single agent
- **88%** of Grade 3-4 events outside the Cycle 1 DLT window[8]

> **What MTD-only dose-finding was never built to handle, all at once:** joint surface unsearched, non-additive toxicity, late-onset toxicity.

Why MTD-only dose-finding doesn't fit targeted agents and immunotherapies



The cost of MTD-only dose-finding

- **Sotorasib** (KRAS G12C, first-in-class, approved 2021)[10]:
  - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
  - FDA required a **240 mg vs 960 mg** comparison. **PFS was similar** at one-quarter the exposure[11]
- **MTD ≠ optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions[12]

Same drug class as daraxonrasib.

Project Optimus: the regulatory ground has shifted

How OBD selection actually works

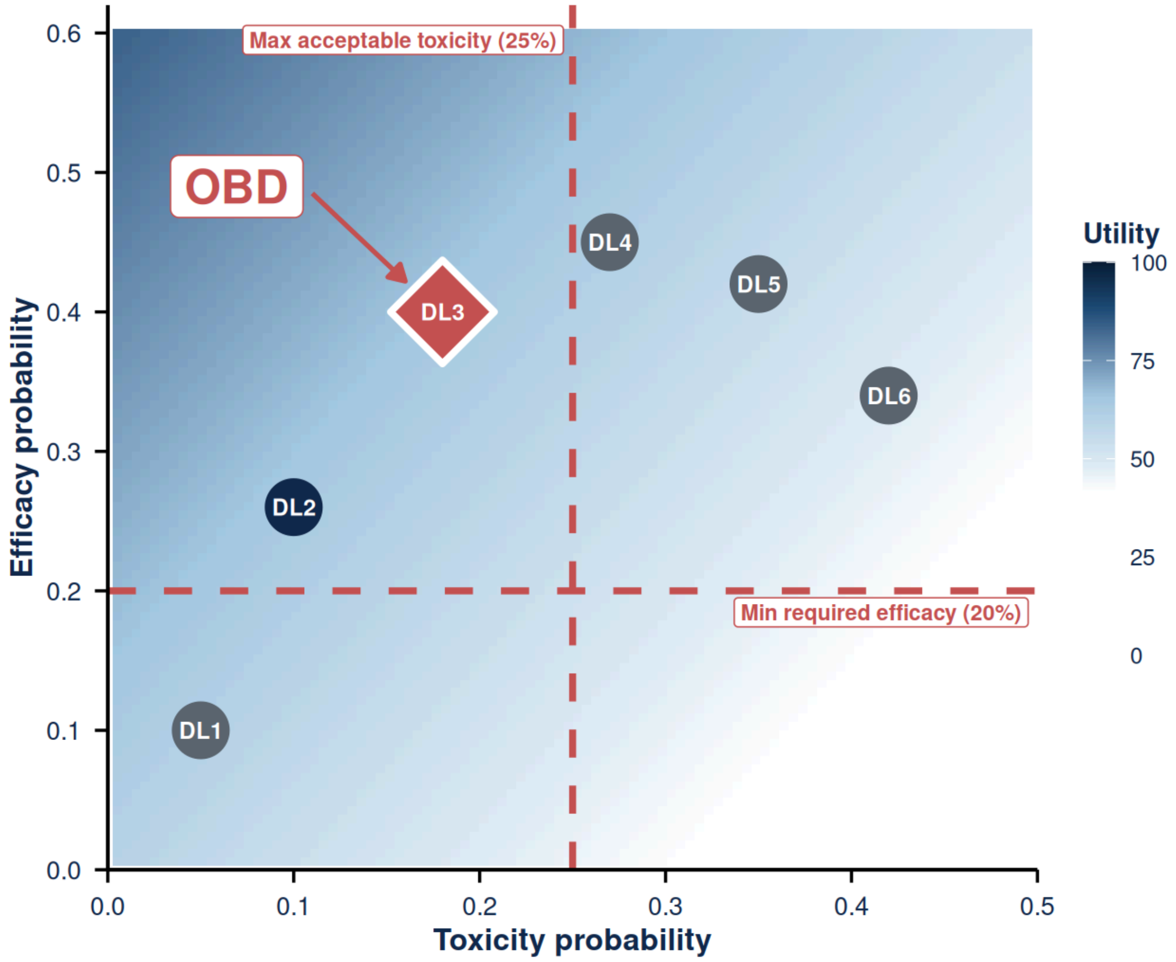
Instead of escalating until toxicity, choose the dose with the best benefit-risk profile.

"A response is worth this; a Grade 3 toxicity costs this."

The team writes that utility table down up front. The design then searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

Elicited utility table (Zhou 2019 framework)[14]:

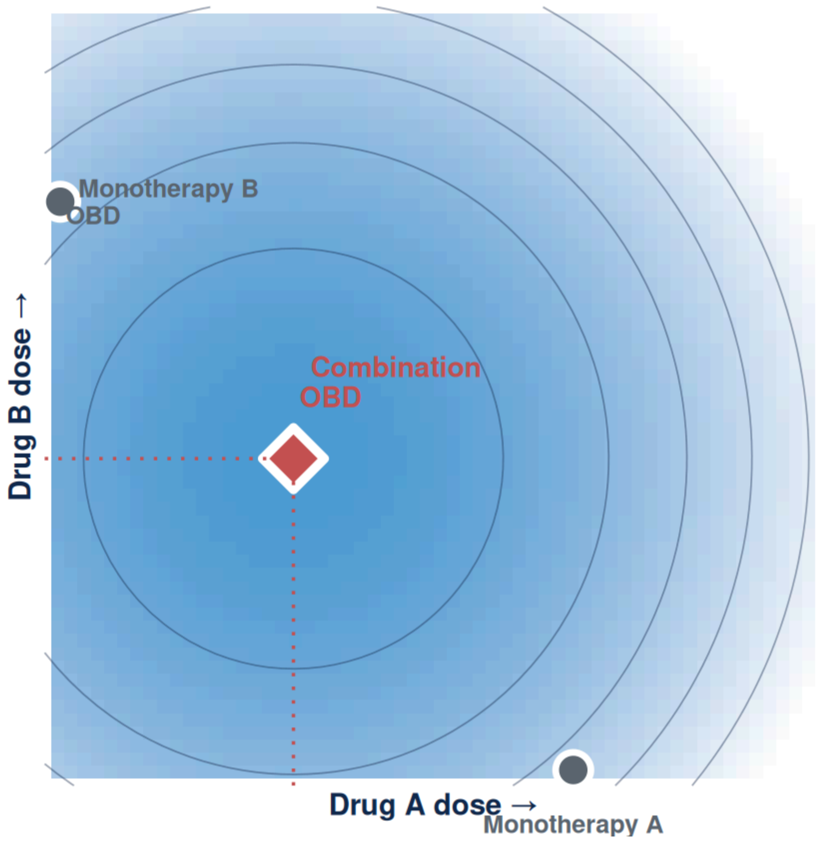
	TOX	NO TOX
Response	40	**100**
No response	**0**	60



***OBD** = admissible dose (inside red lines) with the highest utility.*

Combination dose selection in RAS is harder

Combinations make MTD-only dose-finding worse on every axis.



Conceptual schematic of the benefit-risk surface. The combination OBD sits below both single-agent OBDs; single-agent data alone cannot reconstruct the contour.

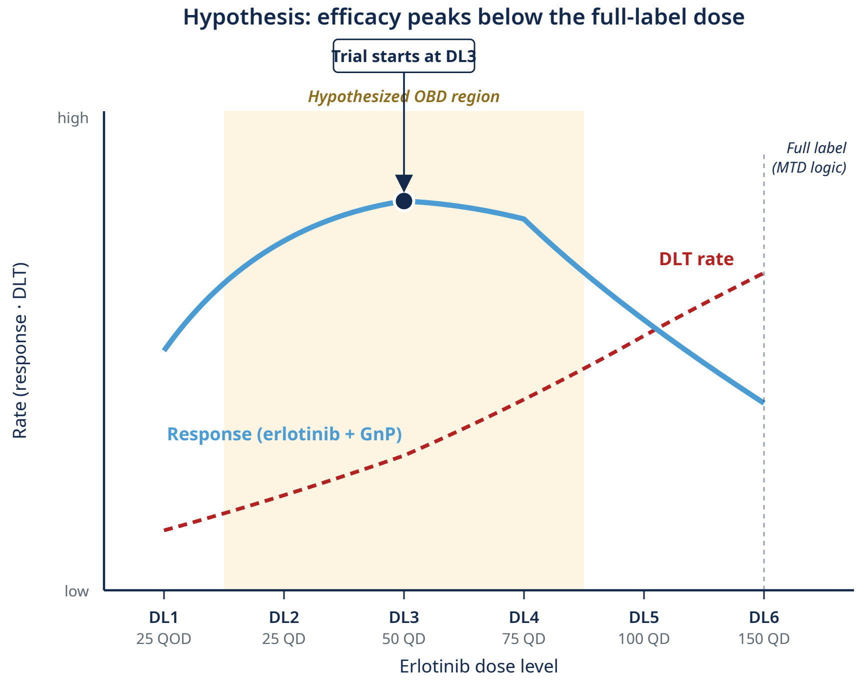
- The combination OBD can sit **below both single-agent OBDs**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **toxicity margin is already spent** at the monotherapy dose

> **Fix what you know. Search what you don't.**

RMC-GI-102 (daraxonrasib + GnP) fixed daraxonrasib at 200 mg without searching the joint surface[4].

Finding the right erlotinib dose in basal-like PDAC

- **Clinical question.** Could *lower*-dose erlotinib work *better* than full-dose erlotinib in basal-like PDAC, when combined with GnP?
- **Design.** First find the safe dose range, then choose the dose with the best response/toxicity trade-off[14,17].
- **Why it matters.** This is an OBD-below-MTD question; 3+3 cannot answer it.

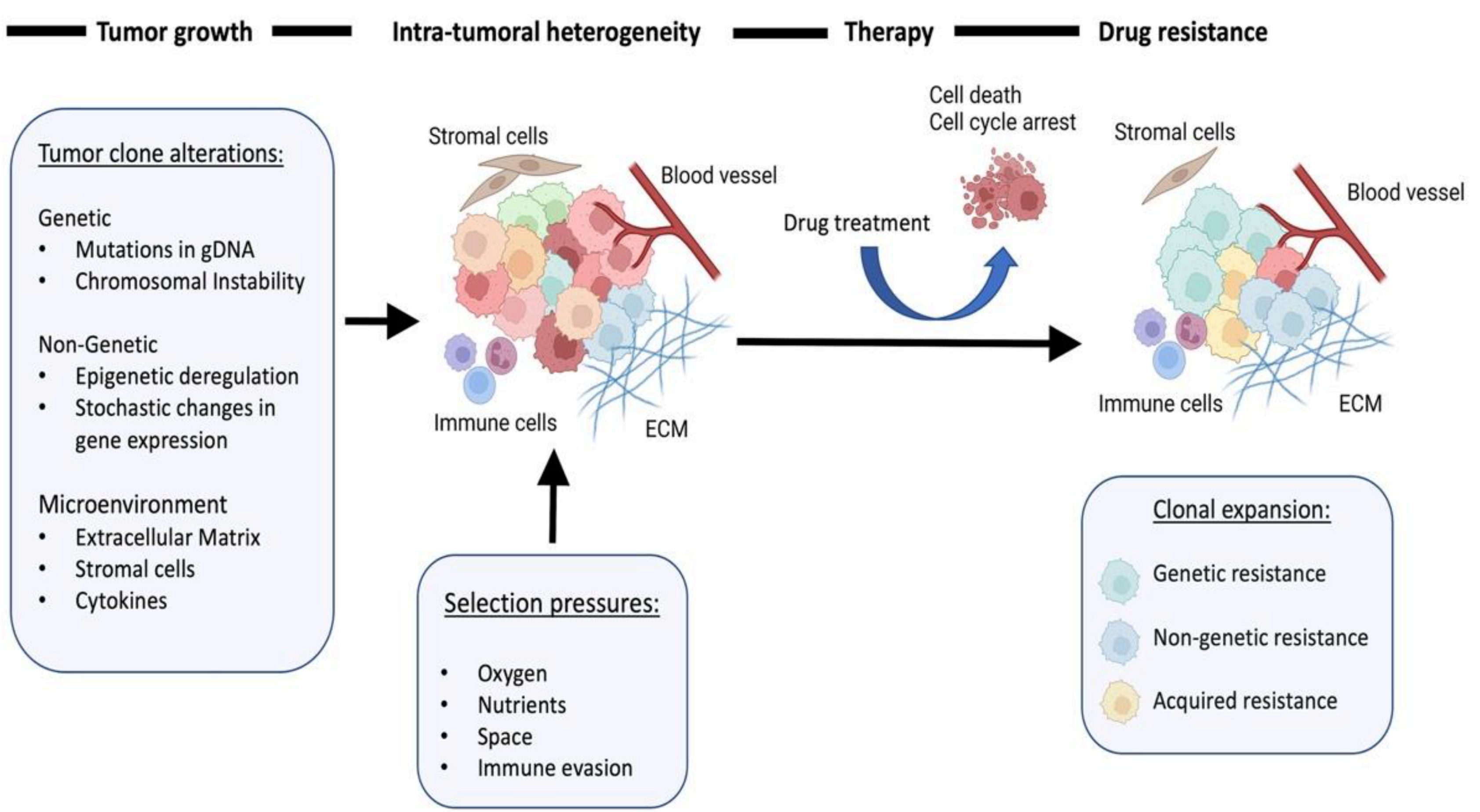


Trial: LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by PurIST[18] · combination erlotinib + biweekly GnP · OBD-below-MTD hypothesized from prior data[19] · **up to 52 patients; ~28 expected at the OBD.**

Resistance is the next problem

Phase 2 onward: *each patient is a moving target*

Resistance is a dynamic process

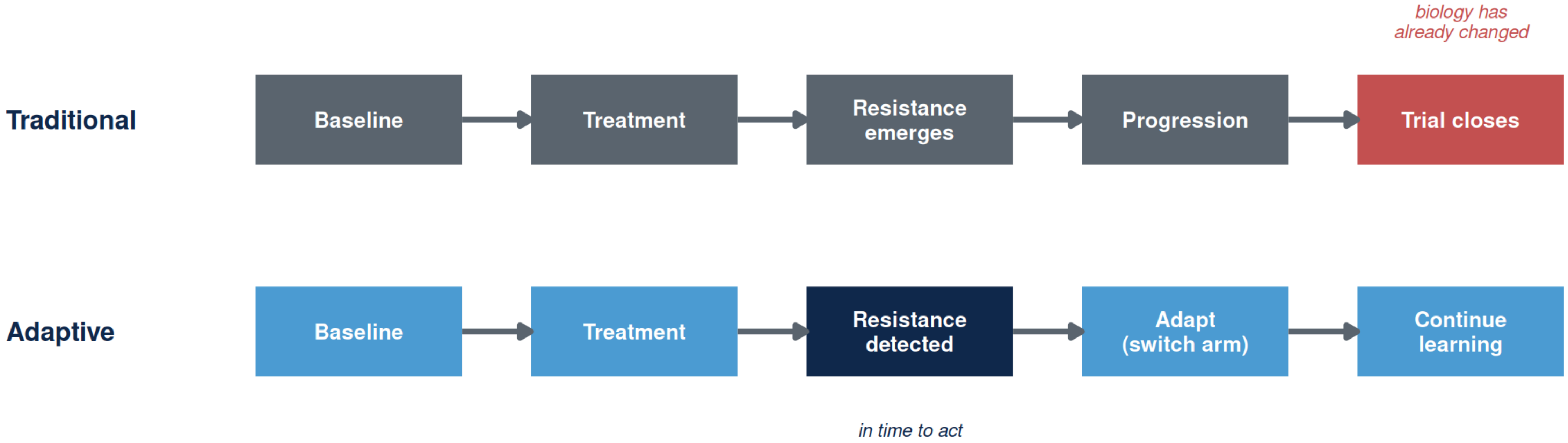


Adapted from Karagiannis & Rampias, Cancers 2022[20].

Resistance mechanisms emerge under RAS-inhibitor pressure

> **Daraxonrasib monotherapy in mPDAC** (n=44, paired ctDNA): **59% show RAS-pathway resistance at progression — KRAS amplification in 36%**[21]

Why classical trial designs miss resistance



RESISTANCE TYPE	WHAT YOU NEED
Genomic (KRAS amplification, NRAS/BRAF/MAP2K1)	ctDNA
Cell-state / plasticity (basal↔classical, partial EMT)	biopsy + transcriptomics + imaging

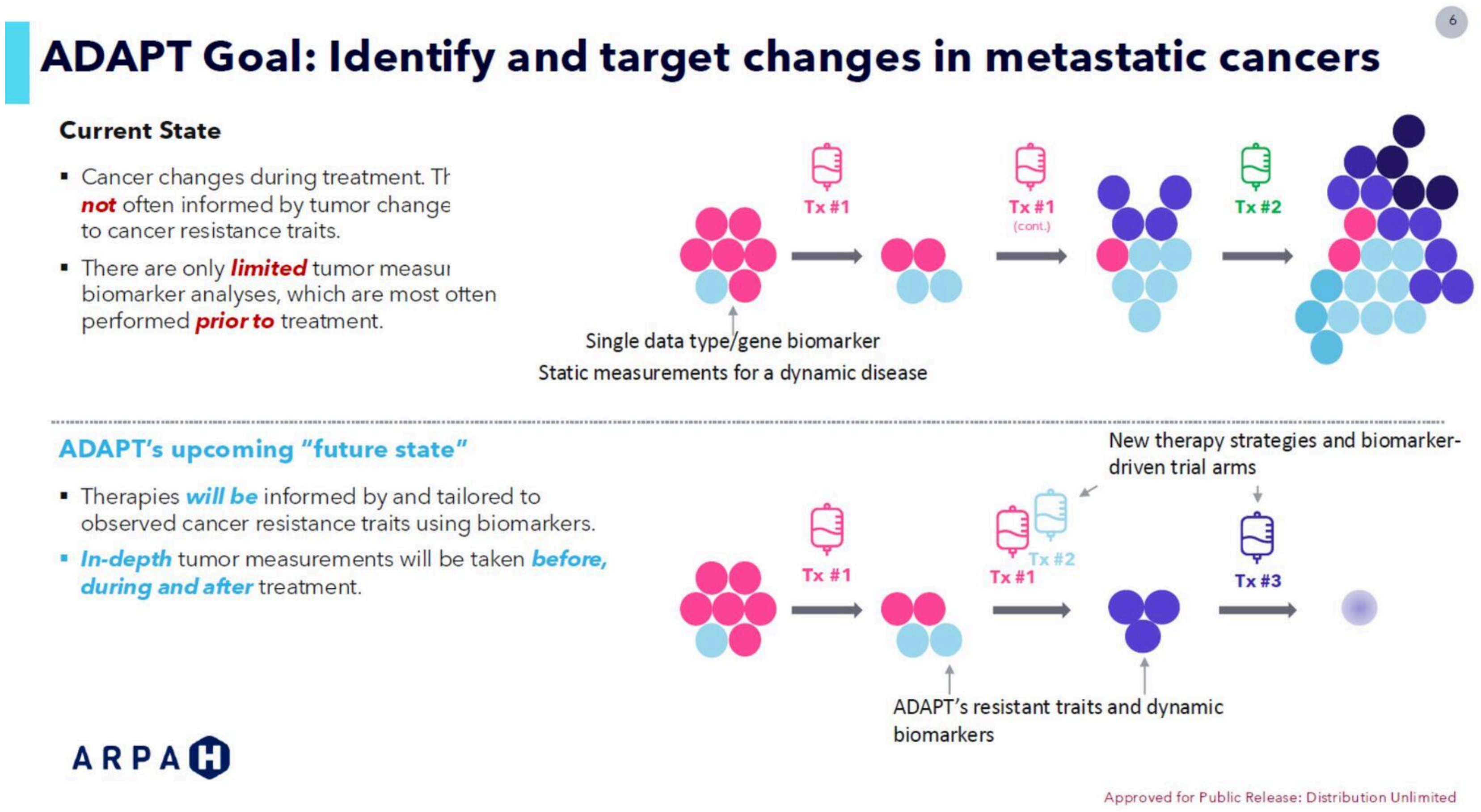
A resistance-aware trial has to **do both** during enrollment[22,23].

ADAPT / EVOLVE: what a resistance-aware trial looks like

ARPA-H's resistance-tracking template.

- Longitudinal resistance detection
- Biomarker-guided adaptation
- Adaptive arm-swaps on emerging biology

*ARPA-H **ADAPT** → **EVOLVE** (UNC-led; Carey MPI, Rashid MPI). **Enrolling.***



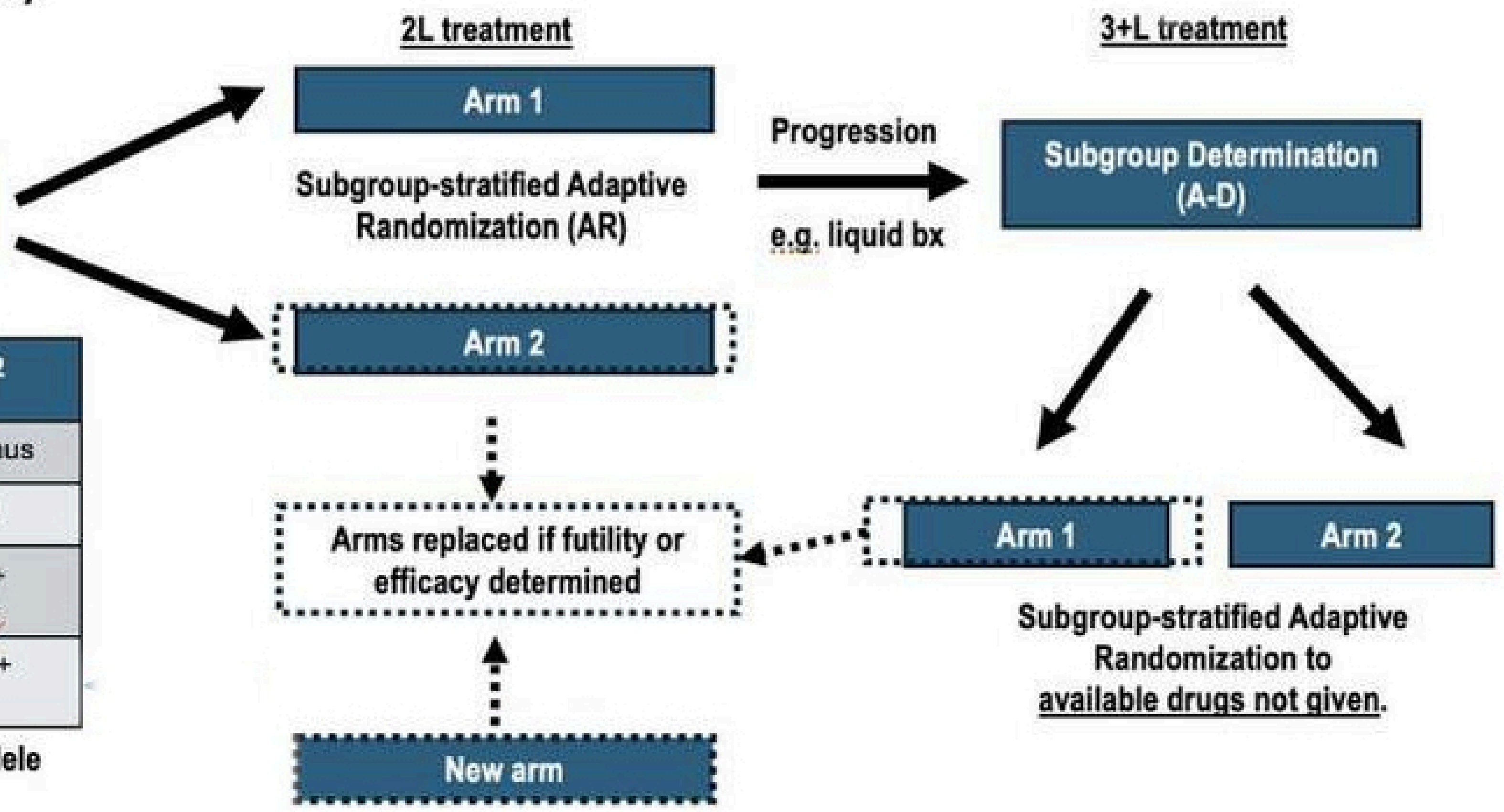
ARPA-H ADAPT — current state (top) vs dynamic-biomarker state (bottom).

EVOLVE-BDT — and what this would look like in PDAC

EVOLVE is the operational template; PDAC is the next application.

First generation trials (pre-TA1 novel biomarkers)

Second line MBC Pts → Biopsy → Subgroup Determination (A-D)



ER+HER2- Initial molecular subgroups* and treatment candidates

Subgroup	ESR1 Mut	PI3K/AKT/PTEN Mt	Initial Arm 1	Initial Arm 2
A	-	-	FUL + CDK4/6i	FUL + Everolimus
B	-	+	FUL + CDK4/6i	FUL + CAPI
C	+	-	Elacestrant + CDK4/6i	Elacestrant+ Everolimus
D	+	+	Elacestrant + CDK4/6i	Elacestrant + CAPI

* Other molecular considerations may include intrinsic subtype by CNV, B-allele frequency, and RB-LOH.

TBCRC EVOLVE-BDT flow — biomarker-stratified subtrials, adaptive randomization, in-trial arm replacement on futility or efficacy.

RASHID LAB
CANCER DATA SCIENCES

EVOLVE-BDT (real, enrolling)

ER+/HER2– and TNBC, biomarker-stratified subtrials with Bayesian arm-swap on futility/efficacy.

PDAC analog (hypothetical)

Basal-like vs classical by PurIST[18], same Bayesian arm-swap logic on daraxonrasib-combination arms.

> **Same architecture, different disease.**

What this demands of the statistical design

Four design axes any such trial needs.

Adaptive stopping	Real-time biomarker discovery	Longitudinal monitoring	In-trial evaluation
Bayesian futility / efficacy boundaries at each interim	Pre-specified signatures + agnostic discovery window	Serial ctDNA, imaging, biopsies on every patient	Biomarker-guided cohorts within the same protocol

Methods are ready. We’re operationalizing them in PDAC trials — starting with PANGEA.

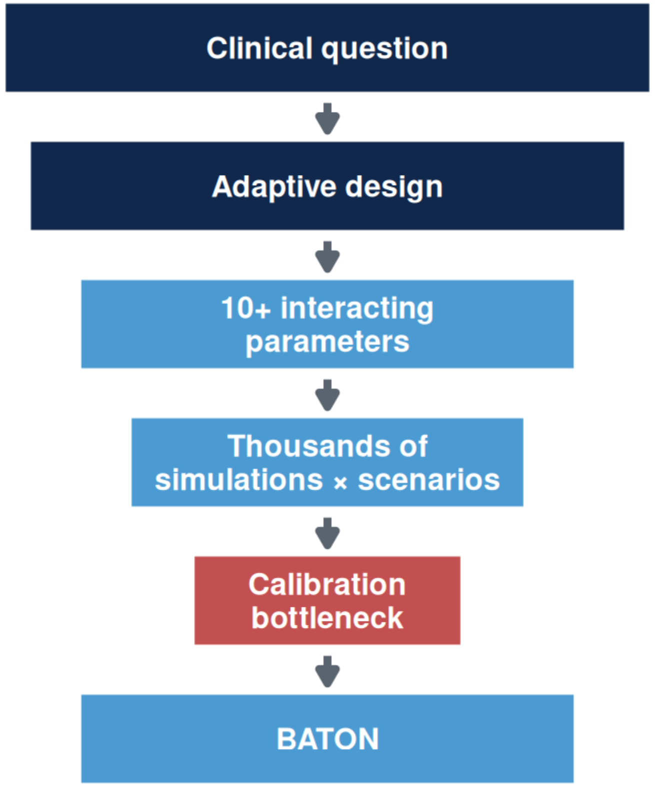
Putting these into practice

Why don’t we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.

Calibration tunes a design to hit its ***operating characteristics*** *— the trial’s report card: false-positive control, power, expected enrollment, early stopping, and time to readout.*



Methods adopted. Dose-optimization plans — where calibration cost bites — not[24].

Our group: several months per design. For combined dose + resistance designs, hand-tuning is infeasible.

The calibration challenge: four competing goals

Designing an adaptive trial requires achieving all four simultaneously.

1. FDA compliance

Power ≥ 0.80
Type I ≤ 0.10

2. Minimize patients

Smaller E[N]
Efficient use of scarce patients

3. Stop bad early

Futility stopping
Protect patients from inactive arms

4. Accelerate good

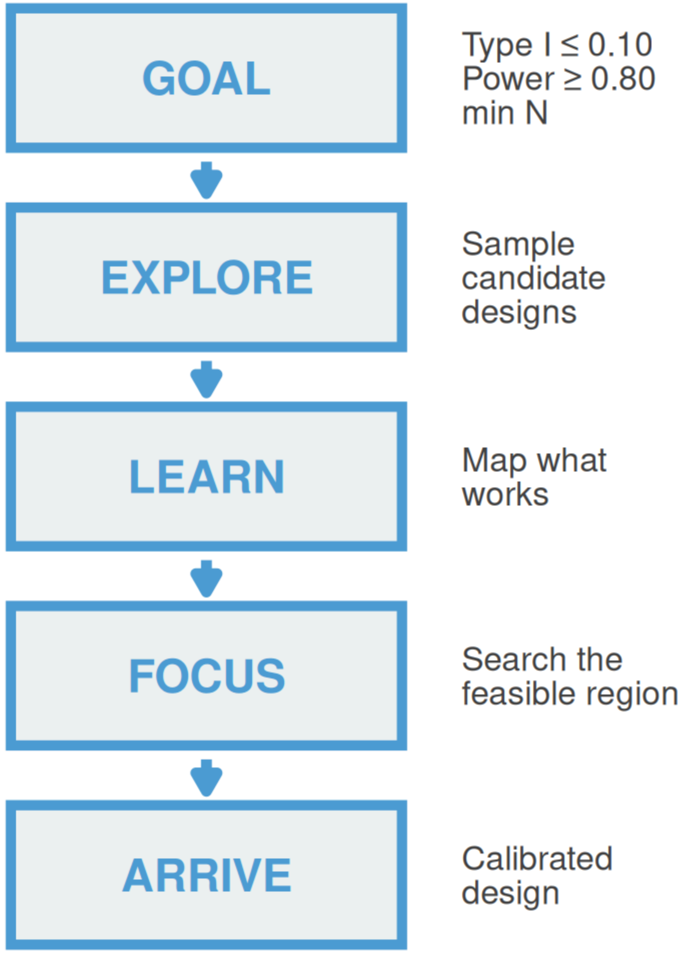
Efficacy stopping
Faster access when drug works

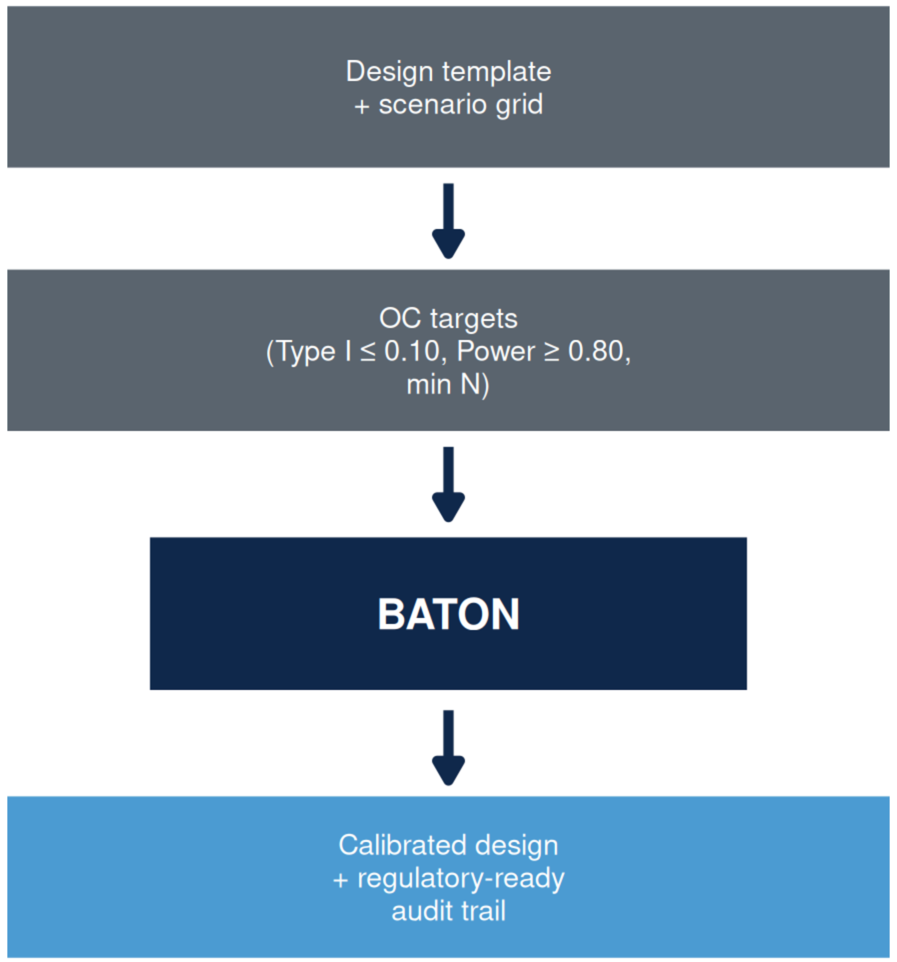
"We could design the science. We needed a faster way to calibrate the trial."

BATON: the calibration tool

BATON = **B**ayesian **A**daptive **T**rial **O**ptimizatio**N**[25]

BATON turns clinically necessary trial complexity into calibrated, launchable designs.





You bring the clinical inputs; BATON returns the calibration.

You bring the clinical design; BATON evaluates hundreds of candidates by simulation and returns the ones that meet your targets. R package: github.com/naimurashid/BATON[25]

A BATON-calibrated design hits all four goals — for PANGEA Phase II

1. FDA compliance ✓

Type I 7.0% (target ≤10%)
Power 82.2% (target ≥80%)

2. Minimize patients ✓

~23 patients spared
(avg N 53 vs Max N 76)

3. Stop bad early ✓

78% early futility stop
when drug is inactive

4. Accelerate good ✓

81% early efficacy stop
when drug works

Far fewer evaluations than grid search — and BATON converges to the optimum where it exists. Manual calibration: days. BATON: minutes, with a regulatory-ready audit trail.

What this means for a PDAC patient

Mr. Hernandez, 62, mPDAC, basal-like by PurIST, progressed on 1L GnP.

Traditional fixed Phase II

- All **76 patients** enrolled before any look
- Drug effect discovered only at study end
- **~36+ months** to read out

PANGEA Phase II (BATON-calibrated)

- **78% chance** of early futility stop **when the drug is inactive**
- Average patients enrolled: **~53** if inactive / **~58** if active
- **27.3 months** expected duration

Sooner answer, fewer patients on a losing arm — without losing power.

PANGEA Phase II: calibrated by BATON

BATON identified a design that preserves power, controls false positives, and allows early stopping when erlotinib isn't helping.

- After Phase I OBD → **Phase II Bayesian adaptive RCT:** GnP + erlotinib (at OBD) vs GnP alone, primary PFS
- **Design assumes** reference 5.5 mo → 9.2 mo (HR 0.58)[26]
- **BATON-calibrated** four parameters under Type I ≤ 0.10, Power ≥ 0.80[25]

Achieved operating characteristics:

METRIC	VALUE
Power	**82.2%**
Type I error	7.0%
Max N	**76** (82 accrual w/ buffer)
Expected N under H_0	53.0
Expected N under H_1	58.1
Pr(early futility stop | H_0)	**77.9%**
Pr(early efficacy stop | H_1)	**81.4%**
Expected duration	27.3 mo

H_0 = drug inactive (no PFS benefit over GnP alone) · H_1 = drug active at the assumed effect size (HR 0.58)

Read as: 78% chance of stopping early when the drug is inactive; 81% chance of stopping early when it works.

Without BATON: four interacting parameters rarely hit both constraints simultaneously. **With BATON:** joint calibration against both constraints, ~200 evaluations.

An unexpected finding: *caps set too low suppress early-efficacy declarations — even when the drug works, the trial waits for the final analysis to say so. BATON returned the* ***best trade-off curve*** *across candidate caps; we chose 76 to preserve early-efficacy stopping. Manual calibration would have anchored on the first feasible design.*

Where this is going

BATON calibrates **aggressive**, **conservative**, or **balanced** designs — and ports across GI applications:

PDAC RAS-backbone combinations OBD selection	**CRC** Joint-surface novel-novel doublets
Basket / umbrella Biomarker-stratified subtrials shared decision rules	**Robust calibration** Designs robust to uncertain trial assumptions

BATON is the infrastructure that makes adaptive trial design routine across the GI portfolio.

Take-home

1. **The RAS era needs new kinds of trials** — biomarker-guided, dose-optimized, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs can handle dose optimization, biomarker guidance, and resistance-aware adaptation.
3. **BATON makes them practical to calibrate and launch.**

Acknowledgments

Clinical collaborators

- Lisa Carey (UNC, EVOLVE PI)
- Channing Der (UNC, NEJM editorial[27])
- Jen Jen Yeh (UNC Surgery, SToP SPORE PI · NEJM editorial · PDAC collaborator)
- Ashwin Somasundaram (PANGEA PI)

EVOLVE MPI team

- Charles Perou (UNC) · Ian Krop, Eric Winer (Yale) · Antonio Wolff (Hopkins)

Methods collaborators

- Stephen Bates (MIT) · Ying Yuan (MD Anderson) · Qian Cao (FDA CDRH)

EVOLVE Statistical Working Group

- Susan Hilsenbeck (Baylor) · Nabihah Tayob (DFCI) · Ruizhe Chen (Hopkins)

Rashid Lab

- **Amber Young**, Tyler Humpherys, Dinelka Nanayakkara, Jialiu Xie, Andrew Walther

Funding

- **NCI P50-CA257911** — UNC **SToP** SPORE
- ARPA-H 140D042590009
- DOD HT9425241103100121
- NCI U01-CA274298

References

1. Eileen M. O'Reilly, Zev A. Wainberg, Andrew E. Hendifar, Mitesh J. Borad, Filippo Pietrantonio, Shubham Pant, Pascal Hammel, Chiara Cremolini, Gulam A. Manji, Paul E. Oberstein, Ignacio Garrido-Laguna, Christoph Springfeld, Nilofer S. Azad, Makoto Ueno, Stephen Y. Chui, Ying Zhang, Hina Patel, Yeonju Lee, Zeena Salman, Brian M. Wolpin, RASolute 302 Trial Investigators. Daraxonrasib or chemotherapy in previously treated metastatic pancreatic cancer. New England Journal of Medicine. 2026. doi:10.1056/NEJMoa2605555
2. Brian M. Wolpin, others. Daraxonrasib in previously treated advanced RAS-mutated pancreatic cancer. New England Journal of Medicine. 2026;394(18):1790–802. doi:10.1056/NEJMoa2505783
3. Yongxian Zhuang, others. The RAS(ON) multi-selective inhibitor daraxonrasib (RMC-6236) induces receptor tyrosine kinase cell-surface expression on pancreatic cancer cells. Cancer Research. 2026;86(5_Suppl_1):PR009. doi:10.1158/1538-7445.RASONCOTHER26-PR009
4. Brian M. Wolpin, Benjamin L. Musher, Gulam A. Manji, others. Daraxonrasib plus chemotherapy as first-line treatment for patients with metastatic pancreatic adenocarcinoma. AACR Annual Meeting 2026, Abstr LB407 (RMC-GI-102, NCT06445062); 2026.
5. Marie-Karelle Rivière, Christophe Le Tourneau, Xavier Paoletti, Frédéric Dubois, Sarah Zohar. Designs of drug-combination phase I trials in oncology: A systematic review of the literature. Annals of Oncology. 2015;26(4):669–74. doi:10.1093/annonc/mdu516

6. Jennifer A. Harrington, Graham M. Wheeler, Michael J. Sweeting, Adrian P. Mander, Duncan I. Jodrell. Adaptive designs for dual-agent phase I dose-escalation studies. Nature Reviews Clinical Oncology. 2013;10(5):277–88. doi:10.1038/nrclinonc.2013.35
7. Christophe Le Tourneau, J. Jack Lee, Lillian L. Siu. Dose escalation methods in phase I cancer clinical trials. Journal of the National Cancer Institute. 2009;101(10):708–20. doi:10.1093/jnci/djp079
8. Bob T. Li, Gerald S. Falchook, Gregory A. Durm, others. OA03.06 CodeBreaK 100/101: First report of safety/efficacy of sotorasib in combination with pembrolizumab or atezolizumab in advanced KRAS p.G12C NSCLC. Journal of Thoracic Oncology. 2022;17(9):S10–1.
9. Amita Patnaik, others. Phase I study of pembrolizumab (MK-3475; anti–PD-1 monoclonal antibody) in patients with advanced solid tumors. Clinical Cancer Research. 2015;21(19):4286–93. doi:10.1158/1078-0432.CCR-14-2607
10. David S. Hong, others. $KRAS^{G12C}$ inhibition with sotorasib in advanced solid tumors. New England Journal of Medicine. 2020;383(13):1207–17. doi:10.1056/NEJMoa1917239
11. Maximilian J. Hochmair, Karim Vermaelen, Giannis Mountzios, others. Sotorasib (960 mg or 240 mg) once daily in patients with previously treated KRAS G12C-mutated advanced NSCLC. European Journal of Cancer. 2024;208:114204. doi:10.1016/j.ejca.2024.114204
12. Hao Zhu, Alex N. Phipps, Ying Yuan, Brad A. Davidson, Stacy S. Shord, Jiang Liu, Patricia M. LoRusso. FDA–AACR strategies for optimizing dosages for oncology drug products: Selecting dosages for first-in-human trials. Clinical Cancer Research. 2025;31(23):4872–81. doi:10.1158/1078-0432.CCR-25-0095

13. U.S. Food and Drug Administration. Optimizing the dosage of human prescription drugs and biological products for the treatment of oncologic diseases. Guidance for Industry (final); 2024.
14. Yanhong Zhou, J. Jack Lee, Ying Yuan. A utility-based bayesian optimal interval (U-BOIN) phase I/II design to identify the optimal biological dose for targeted and immune therapies. Statistics in Medicine. 2019;38(28):5299–316.
15. Ruitao Lin, Yanhong Zhou, Fangrong Yan, Daniel Li, Ying Yuan. BOIN12: Bayesian optimal interval phase I/II trial design for utility-based dose finding in immunotherapy and targeted therapies. JCO Precision Oncology. 2020;4:1393–402.
16. Peter F. Thall, Elizabeth Garrett-Mayer, Nolan A. Wages, Susan Halabi, Ying Kuen Cheung. Current issues in dose-finding designs: A response to the US FDA oncology center of excellence project optimus. Clinical Trials. 2024;21(3):267–72.
17. Suyu Liu, Ying Yuan. Bayesian optimal interval designs for phase I clinical trials. Journal of the Royal Statistical Society: Series C (Applied Statistics). 2015;64(3):507–23.
18. Naim U. Rashid, Xianlu L. Peng, Chong Jin, Richard A. Moffitt, Keith E. Volmar, Brian A. Belt, Roheena Z. Panni, Timothy M. Nywening, Stephen G. Herrera, Kathleen J. Moore, Sharon G. Hennessey, Ashley B. Morrison, Ryan Kawalerski, Apoorve Nayyar, Albert E. Chang, Bradley Schmidt, Hong Jin Kim, David C. Linehan, Jen Jen Yeh. Purity independent subtyping of tumors (PurIST): A clinically robust, single-sample classifier for tumor subtyping in pancreatic cancer. Clinical Cancer Research. 2020;26(1):82–92. doi:10.1158/1078-0432.CCR-19-1467
19. Steven J. Cohen, others. A phase 1b study of erlotinib in combination with gemcitabine and nab-paclitaxel in patients with previously untreated advanced pancreatic cancer. Cancer

Chemotherapy and Pharmacology. 2016;77(4):693–701.
20. Dimitris Karagiannis, Theodoros Rampias. Cancer evolution in precision medicine era. Cancers (Basel). 2022;14(8):1885. doi:10.3390/cancers14081885
21. Ida Aronchik, others. Mechanisms of resistance to the RAS(ON) multi-selective inhibitor daraxonrasib in pancreatic cancer. ENA Symposium 2025, Revolution Medicines poster; 2025.
22. Mark M. Awad, others. Acquired resistance to KRAS G12C inhibition in cancer. New England Journal of Medicine. 2021;384(25):2382–93. doi:10.1056/NEJMoa2105281
23. others. Mechanisms of resistance to oncogenic KRAS inhibition in pancreatic cancer. Cancer Discovery. 2024;14(11):2135–61. doi:10.1158/2159-8290.CD-24-0177
24. Deepak Bhamidipati, others. Impact of FDA project optimus guidance on the design of early-phase clinical trials. JCO Oncology Advances. 2025. doi:10.1200/OA-25-00084
25. Amber M. Young, others. BATON: Constrained Bayesian optimization for calibrating adaptive trials. 2026.
26. Daniel D. Von Hoff, others. Increased survival in pancreatic cancer with nab-paclitaxel plus gemcitabine. New England Journal of Medicine. 2013;369(18):1691–703. doi:10.1056/NEJMoa1304369
27. Channing J. Der, Jen Jen Yeh. Advances in RAS therapeutics for pancreatic cancer. New England Journal of Medicine. 2026;394(18):1857–61. doi:10.1056/NEJMe2600517

Questions